

Kyle Webb

Staff Data Scientist – statistics, data engineering, and applied AI
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PROFESSIONAL SUMMARY

Data scientist and data engineer with 10+ years across scientific research, federal health programs, and industry. He builds both sides: the data platforms, pipelines, and governance that make data trustworthy (terabytes across 20 sources, 2,000+ governed users, 82% runtime and 93% cost cuts) and the machine learning and applied statistics that turn it into predictions and decisions.

EXPERIENCE

eSimplicity — Staff Data Scientist

Sept 2023 – Present · Remote

Senior Data Scientist, Apr 2022 – Aug 2023 · AI Data Analyst Fellow, Rapid Innovation Center (internal applied-AI fellowship)

- As AI Data Analyst Fellow, built a Databricks Apps showcase of applied-AI use cases: a cost-analysis dash app over synthetic spend data (users and Unity Catalog resource groups) with live natural-language Q&A (Claude Sonnet 4.6) via an MCP server; evaluated with pytest and automated Playwright tests.
- Led six DevOps and data engineers shipping validated data distributions across 20 CDR sources: terabytes of claims, provider, and beneficiary data orchestrated with AWS MWAA, Terraform, and Databricks.
- Spearheaded migration of all legacy EMR workloads to Databricks, building automated validation, testing, and monitoring, slashing median runtime 82% and median cost 93%.
- Architected data-governance workflows enforcing strict role-based access controls for 2,000+ users across a Kinesis-driven data repository.
- Launched Databricks Delta Shares for governed cross-workspace sharing, eliminating duplicate data copies and their cost.
- Publishes applied side projects with interactive write-ups in a public Mini Project Space (webbkyle.github.io/mini-projects).

GovernmentCIO — Data Scientist

July 2021 – Apr 2022 · Remote

- Single-handedly automated VA IT infrastructure reporting in R/Python as the team's sole developer; briefed major incident data to Operations Triage Group leadership and presented to 200+ attendees including the VA Chief Information Officer.
- Delivered end-to-end analysis (querying, cleaning, modeling) for 30+ briefs spanning invalid security certificates to telecom outages for the largest U.S. healthcare network, several on one-day turnarounds.

National Human Genome Research Institute (NHGRI) — Bioinformatics Scientist

Oct 2020 – July 2021 · Bethesda, MD

- Stood up the lab's N3C data-enclave workspace from scratch; secured data use agreements for COVID-19 EHR pipelines and coached teammates on PySpark/SparkR.
- Curated and analyzed EHR data from millions of patients powering a COVID-19 Phenotype-Wide Association Study (PheWAS).
- Harmonized N3C and All of Us enclave data for a participant comparison study, applying stratified sampling across race, gender, age, and ethnicity to isolate true COVID-19 effects.

National Institute of Allergy and Infectious Diseases (NIAID) — Data Science Fellow

June 2018 – Sept 2020 · Bethesda, MD

- Engineered ETL pipelines and parallelized HPC jobs over 4TB+ of Cerner data to classify rare diseases with machine learning.
- Rebuilt performance and network visualization for SIGNAL, a public genetic screening tool (R Shiny + D3.js), shipping in four months a roadmap scoped for six.
- De-identified 32,000 clinical notes, then trained and benchmarked two NLP models for Human Phenotype Ontology assignment, visualizing results with hierarchical t-SNE renderings.
- Extended the open-source R package *ihazr*, adding semiparametric hazard regression estimation and visualization for survival analysis.
- Completed five cross-institute data-science rotations: biostatistics, immunology, bioinformatics, grants analytics, epidemiology (detailed on CV).

SELECTED PROJECTS

30-day hospital readmission risk: production-style ML system with FHIR-native pipeline, deployed LightGBM model, drift monitoring, weekly retraining ([demo](#) · [write-up](#)) ... **How many fraudulent Social Security beneficiaries are there?** Bayesian estimate from public SSA records, interactive posteriors ([write-up](#))

EDUCATION

- **M.S., Statistics** — Virginia Tech (2015 – 2016)
- **B.S., Statistics** — Wake Forest University (2011 – 2015), GPA 3.4/4.0

CERTIFICATIONS

AWS Certified Cloud Practitioner (2023) · AWS Certified Solutions Architect – Associate (2026) · Databricks Certified Data Engineering Associate (2026)

TECHNICAL SKILLS

Python · R · SQL · Linux · Git · Terraform · AWS · Databricks · Spark/PySpark · Airflow (AWS MWAA) · dbt · Docker · MLflow · scikit-learn · LLMs